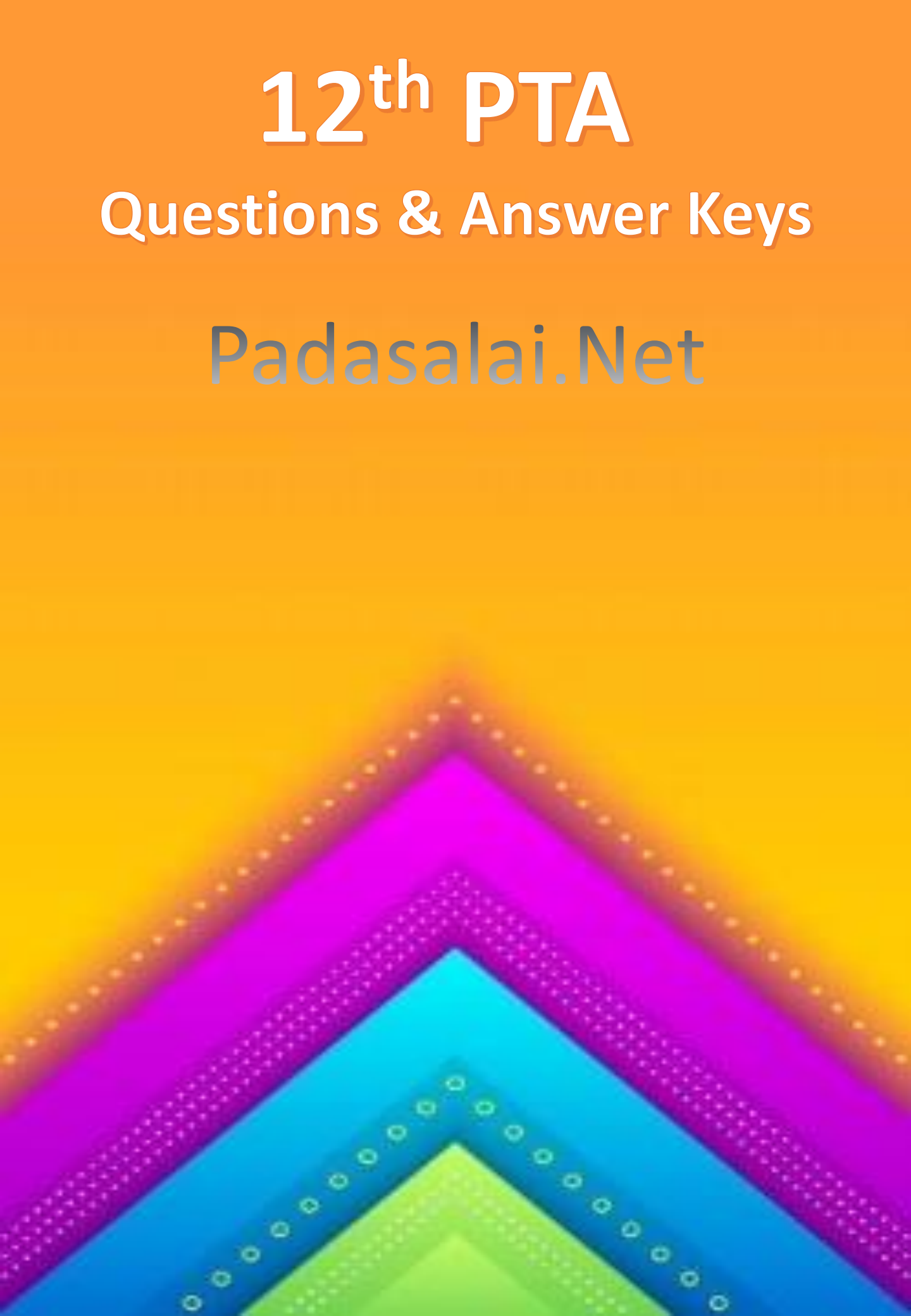
12th PTA

Questions & Answer Keys

Padasalai.Net



Padasalai.Net CBSEtips.in

PHYSICS

PTA - MODEL QUESTION PAPER - 5
STANDARD XII

TIME ALLOWED : 15min + 3.00 Hr **Maximum Marks : 70**

PART – I

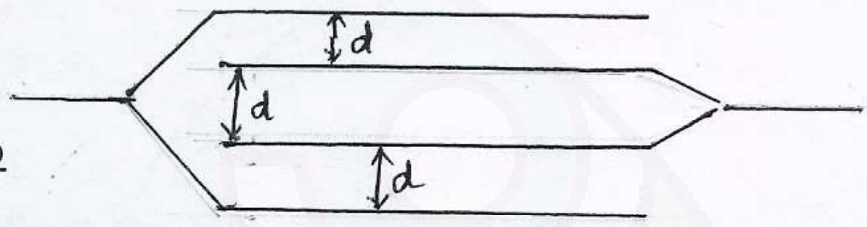
Note: (i) Answer all the questions **15× 1 =15**

(ii) Choose the most appropriate answer from four given alternatives and write the option code with the corresponding answer

1. Five balls marked 1,2,3,4 and 5 are suspended by separate threads. The pairs (1,2)(2,4) and (4,1) show mutual attraction and the pairs (2,3) and (4,5) show repulsion. The nature of ball marked as 1 is

a) positive b) negative **c) neutral** d) can't determine

2. The resultant capacitance of four plates, each is having an area A, arranged as shown above, will be (plate separation is d)



a) $\frac{A\varepsilon_0}{d}$ b) $\frac{A\varepsilon_0}{2d}$

c) $\frac{\mathbf{2A\varepsilon_0}}{\mathbf{d}}$ d) $\frac{3A\varepsilon_0}{d}$

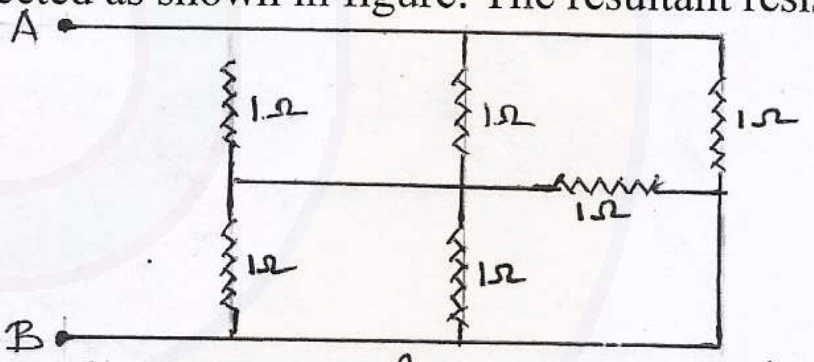
Hints:This four plates form two capacitors, which are parallel. $C = C_1+C_2 = C + C = 2C = \frac{2A\varepsilon_0}{d}$

3. Kirchoff's I and II laws are based on conservation of

a) charge and energy b) energy and charge

c) energy and voltage d) energy and current

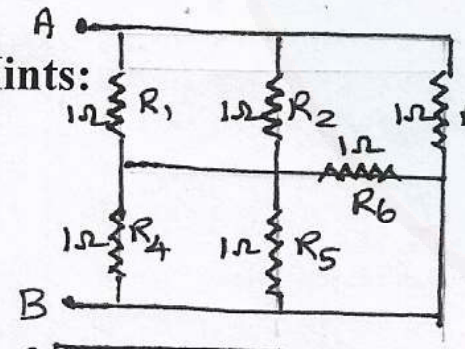
4. The resistors each of resistance 1Ω is connected as shown in figure. The resultant resistance between A and B is



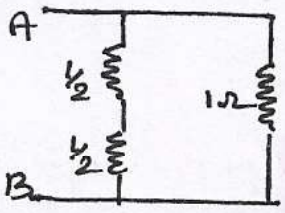
a) $\frac{11}{5}\Omega$ **b)** $\frac{\mathbf{5}}{\mathbf{11}}\mathbf{\Omega}$

c) $\frac{6}{5}\Omega$ d) $\frac{5}{6}\Omega$

Hints:

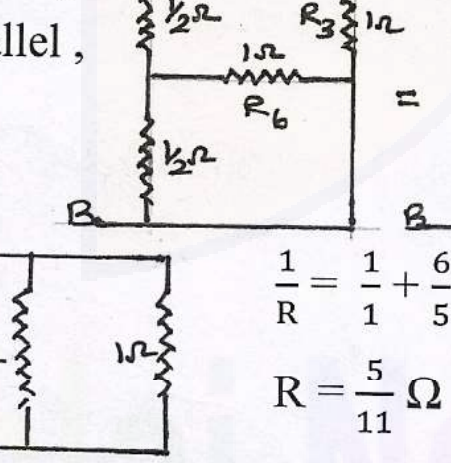


R_1,R_2 are in parallel, R_4,R_5 are in parallel , $\frac{1}{R_p} = \frac{1}{1}+\frac{1}{1} = \frac{2}{1}$, $R_P = \frac{1}{2}\Omega$



$\frac{1}{2}, \frac{1}{3}$ are in series

$R_S = \frac{1}{2}+\frac{1}{3} = \frac{5}{6}\Omega$



R_6, $\frac{1}{2}\Omega$ are in parallel $\frac{1}{R_p} = \frac{1}{1}+\frac{2}{1} = \frac{3}{1}$ $R_P = \frac{1}{3}\Omega$

$\frac{1}{R} = \frac{1}{1}+\frac{6}{5} = \frac{5+6}{5} = \frac{11}{5}$

$R = \frac{5}{11}\Omega$

5. The magnetic field at the centre of a current carrying circular loop is B. If the radius of the loop is doubled, keeping the current as same the magnetic field at the centre of the loop would be

a) B **b)** $\frac{\mathbf{B}}{\mathbf{2}}$ c) $\frac{B}{4}$ d) 2B

Hint: $B = \frac{\mu_0 I}{2a}$; $B' = \frac{\mu_0 I}{2\times 2a} = \frac{B}{2}$

6. A simple pendulum with charged bob is oscillating with time period T and let θ be the angular displacement. If the uniform magnetic field is switched on in a direction perpendicular to the plane of oscillation then

a) time period will decrease but θ will remain constant

b) time period will remain constant but θ will decrease

c) both time period and θ remain constant

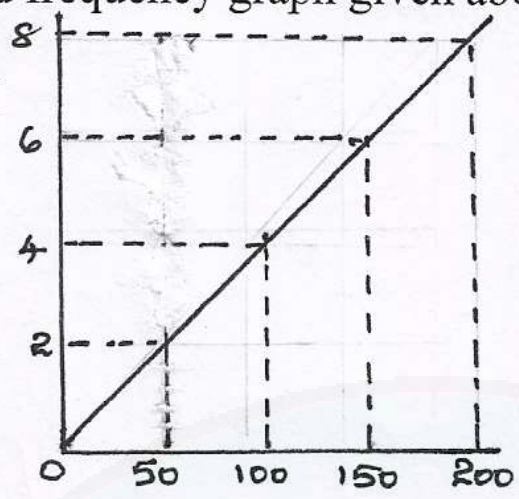
d) both time period and θ will decrease

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

7. If the angular speed of rotation of an armature of AC generator is doubled, the induced emf will be a) same **b) doubled** c) halved d) quadrupled

8. From the reactance and frequency graph given above the value of the inductance of the inductor is



a) 3.18×10^{-14} H

b) $1/100\pi$ H

c) 50π H

d) 6.37×10^{-3}H

Hint: $X_L = 2\pi fL$,

$L = \frac{X_L}{2\pi f} = \frac{2}{2\pi \times 50} = \frac{2}{100\pi}$

$L = 2 \times 10^{-2} \times 0.3185 \quad \because \frac{1}{\pi} = 0.3185$

$L = 6.37 \times 10^{-3}$H

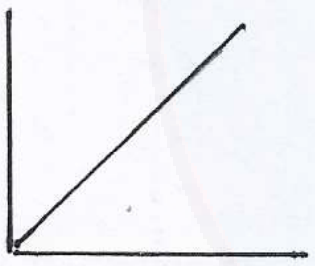
9. Stars twinkle due to:
a) reflection b) total internal reflection
c) refraction d) polarization

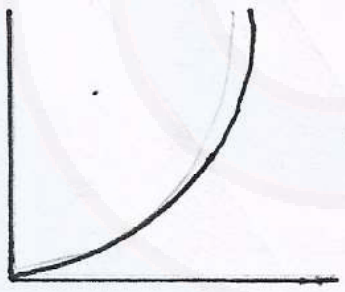
10. The equi-convex lens, has a focal length f. The focal length of each half if the lens cut along AB is a) f/2 b) f c)3f/2 **d) 2f**

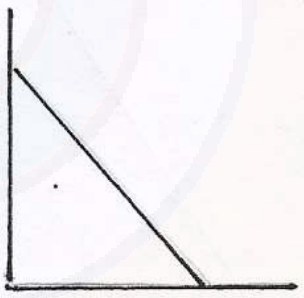
Hint: Before cutting, $R_1 = R$, $R_2 = -R$ $\therefore \frac{1}{f} = (n-1)\left(\frac{1}{R} - \frac{1}{(-R)}\right) = \frac{2(n-1)}{R}$ $\therefore f = \frac{R}{2(n-1)}$

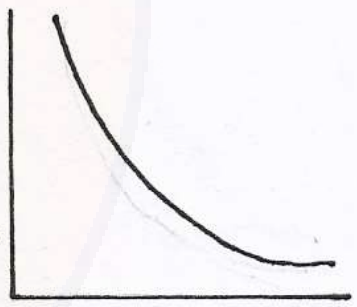
After cutting, $R_1 = R$, $R_2 = \infty$ $\therefore \frac{1}{F} = (n-1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \frac{(n-1)}{R}$ $\therefore F = \frac{R}{(n-1)} = = \frac{2R}{2(n-1)} = 2f$

11. Which of the following graph represents the variation of particle momentum and associated de Broglie wavelength

a) 

b) 

c) 

(d) 

12. If in nuclear fusion process, the masses of the fusing nuclei be m_1 and m_2 and the mass of the resultant is m_3, then
a) $m_3 = m_1 + m_2$ b) $m_3 = m_1 - m_2$ **c) $m_3 < (m_1 + m_2)$** d) $m_3 > (m_1 + m_2)$

13. If a half wave rectified voltage is fed to a load resistor, which part of a cycle the load current will flow?
a) 0° - 90° b) 90° - 180° **c) 0° - 180°** d) 0° - 360°

14. The current gain (β) of a transistor in common emitter mode is 40. To change the collector current by 160mA at constant V_{CE}, the necessary change in the base current is
a) 0.2 mA b) 4μA **c) 4 mA** d) 40 mA

Hint: $\beta = 40$, $I_c = 160$mA $\beta = \frac{I_C}{I_B}$; $I_B = \frac{I_C}{\beta} = \frac{160 \times 10^{-3}}{40} = 4 \times 10^{-3}$A $= 4$ mA

15. Which range of frequencies is suitable for sky wave propagation?
a) 20 KHz to 250 KHz b) 500 KHz to 1500 KHz
c) 10 MHz to 200 MHz **d) 3 MHz to 30 MHz**

PART – II

Answer any six questions . Question number 24 compulsory. **$6 \times 2 = 12$**

16. Can two equipotential surfaces intersect ? Give reason.

No, they can not intersect each other, because two equipotential surface have different electric potential. Si if they intersect then the point of intersection will have two different potential at the same point , which is not possible.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

17. Under what condition, is the heat produced in an electric circuit i) directly proportional and ii) inversely proportional to the resistance of the circuit?

i) Hear produced in the circuit is directly proportional to resistance if a constant current is flowing through a circuit, because $H = I^2Rt$ or $H \propto R$. It is so in series combination of resisters

ii) Hear produced in the circuit is inversely proportional to resistance if a constant potential difference is applied through a circuit, because $H = \frac{V^2}{R} t$ or $H \propto \frac{1}{R}$. It is so in parallel combination of resisters.

18. A parallel combination of two cells of emf's E_1 and E_2 , internal resistance $r_1$and r_2 is used to supply current to a load of resistance R. Write the expression for the current through the load in terms of E_1, E_2 , $r_1$and r_2.

According to Kirchoff's law

$$I = I_1 + I_2$$

Internal resistance (r)

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} \quad (r_1, r_2 \text{ are in parallel})$$

$$\frac{1}{r} = \frac{r_1 + r_2}{r_1 r_2} \quad (\text{or}) \; r = \frac{r_1 r_2}{r_1 + r_2}$$

Net resistance in the circuit $R_{net} = R + \frac{r_1 r_2}{r_1 + r_2}$

$$E = Ir = (I_1 + I_2)\left(\frac{r_1 r_2}{r_1 + r_2}\right) = \left(\frac{E_1}{r_1} + \frac{E_2}{r_2}\right)\left(\frac{r_1 r_2}{r_1 + r_2}\right) = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} \times \frac{r_1 r_2}{r_1 + r_2} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

Current through the load resistance

$$I = \frac{E}{R_{net}} = \frac{\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}}{R + \frac{r_1 r_2}{r_1 + r_2}}$$

$$I = \frac{E_1 r_2 + E_2 r_1}{R(r_1 + r_2) + r_1 r_2}$$

19. Define RMS current. Give the expression.

The root mean square value of an alternating current is defined as he square root of the mean of the squares of all currents over one cycle.

$$I_{RMS} = \frac{I_O}{\sqrt{2}} = 0.707 \, I_0$$

20. What is displacement current?

Displacement current can be defined as the current which comes into play in the region in which the electric field and the electric flux are changing with time

21. What are the differences between interference and diffraction?

S.No.	Interference	Diffraction
1	Equally spaced bright and dark fringes	Central bright is double the size of the other fringes
2	Equal intensity for all bright fringes	Intensity falls rapidly for higher order fringes
3	Large number of fringes are obtained	Less number of fringes are obtained

22. Calculate the momentum and the de Broglie wavelength of an electron with Kinetic energy 25 eV

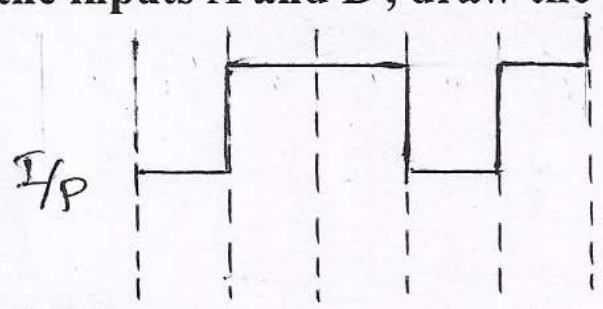
KE = 25 V, ∴ Potential V = 25 V

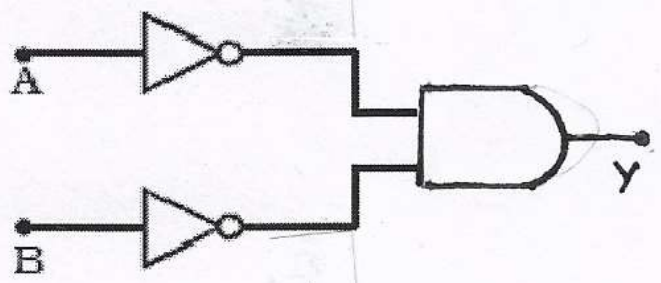
$$\lambda = \frac{12.27\text{Å}}{\sqrt{V}} = \frac{12.27\text{Å}}{\sqrt{25}} = 2.254 \text{ Å}$$

Momentum $p = \frac{h}{\lambda} = \frac{6.626 \times 10^{-34}}{2.454 \times 10^{-10}} = 2.7 \times 10^{-24} \text{ Kgms}^{-1}$

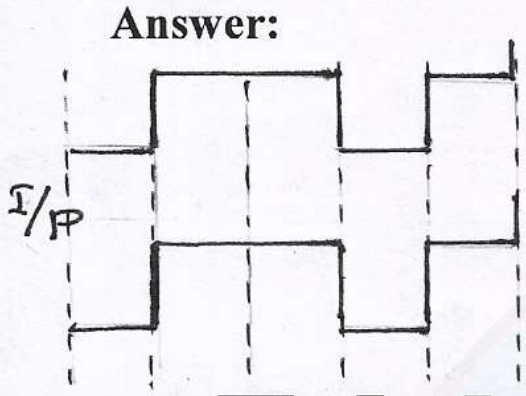
Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

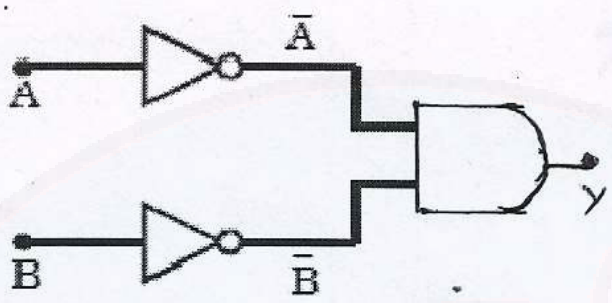
23. Name the logic gate equivalent of the output. If the same waveform I/P is given to both the inputs A and B , draw the corresponding output waveform

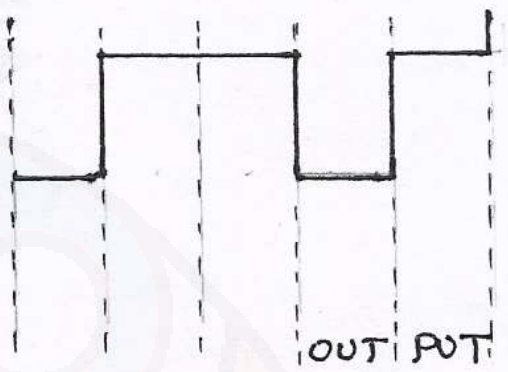




Answer:







$Y = \overline{\overline{A}.\overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$, It is an OR gate.

If the inputs are same waveform I/P(input) is given to both the inputs A and B , the corresponding output is given in the diagram.

24. What is the skip distance and skip zone in Sky wave propagation?

The shortest distance between the transmitter and the point of reception of the sky wave along the surface is called as the skip distance

A zone in between where there is no reception of electromagnetic waves neither ground nor sky, called as skip zone or skip area.

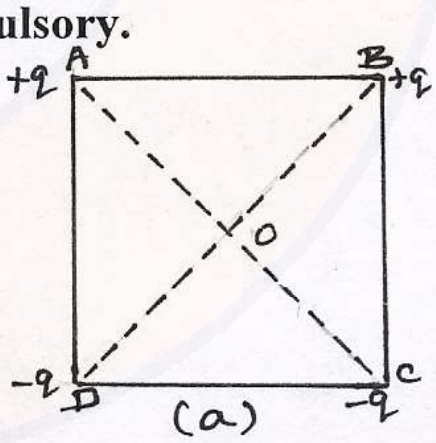
PART - III

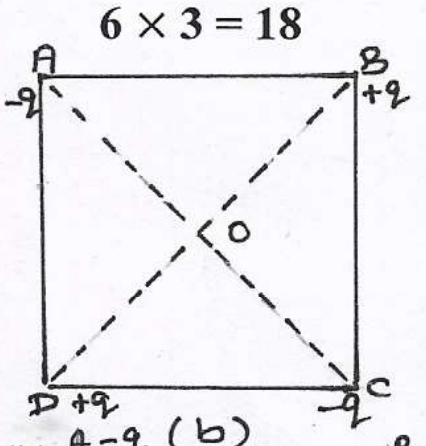
Answer any six questions . Question number 33 compulsory. **6 × 3 = 18**

25. Four point charges are placed at the four corners of a square in two ways (a) and (b) as shown in the figure.

Will the (i) electric potential and

ii) electric field, at the centre of the square be the same or different in two configuration and why?



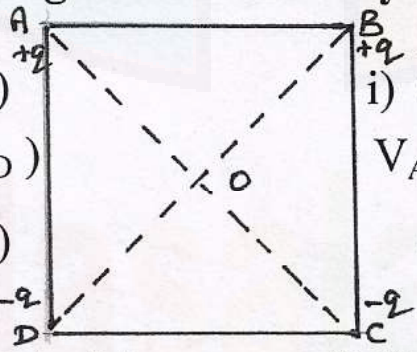


i) electric potential at O in (a)

$V_A = \frac{1}{4\pi\varepsilon_o}\frac{1}{r}(q_A + q_B + q_C + q_D)$

$= \frac{1}{4\pi\varepsilon_o}\frac{1}{r}(+q + q - q - q)$

$= 0$



ii) Resultant electric field at O in (a)

$E_R = \sqrt{(2E)^2 + (2E)^2}$

$= \sqrt{4E^2 + 4E^2}$

$= \sqrt{8E^2}$

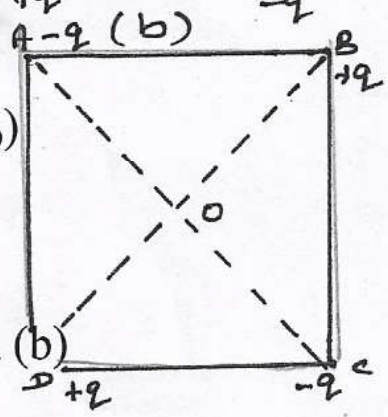
$= 2\sqrt{2}\,E$

i) electric potential at O in (b)

$V_A = \frac{1}{4\pi\varepsilon_o}\frac{1}{r}(q_A + q_B + q_C + q_D)$

$= \frac{1}{4\pi\varepsilon_o}\frac{1}{r}(+q + q - q - q)$

$= 0$



ii) Resultant electric field at O in (b)

$E_A = E_B = E_C = E_D$

E_A , E_C acting in opposite direction and cancel each other

E_B , E_D acting in opposite direction and cancel each other

Net electric field at O is zero, E = 0

In square (a) and (b), potentials are equal , but electric field is different.

Because potential is a scalar, electric field is a vector.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

26. A galvanometer coil of resistance 12Ω gives full scale deflection for 3 mA. How will you convert it into a voltmeter of range 0 to 18 V and ammeter of range 0 to 6A.

$I_g = 3$ mA, $R_g = 12\Omega$, V = 0-18V , I = 0 – 6 A

$$R_h = \frac{V}{R_g} - R_g = \frac{18}{3\times10^{-3}} - 12 = 6000 - 12 = 5988\ \Omega$$

5988 Ω is connected in series with galvanometer.

$$s = \frac{I_g}{I-I_g} \times R_g = \frac{3\times10^{-3}}{(6-3\times10^{-3})} \times 12 = \frac{3\times10^{-3}}{10^{-3}(6000-3)} \times 12 = \frac{3}{5997} \times 12 = \frac{36}{5997} = 0.006\Omega$$

0.006Ω is connected in parallel with galvanometer.

27. Derive an expression for energy stored in an inductor.

Whenever a current is established in the circuit, the inductance opposes the growth of the current. In order to establish a current in the circuit, work is done against this opposition by some external agency. This work done is stored as magnetic potential energy.

Let us assume that electrical resistance of the inductor is negligible and inductor effect alone is considered. The induced emf ε at any instant *t* is

$$\varepsilon = -L\frac{di}{dt}$$

Let dW be work done in moving a charge dq in a time dt against the opposition, then

$$dW = -\varepsilon\, dq$$

$$= -\varepsilon\, idt \quad .....(2) \qquad \because dq = id\,t$$

Substituting for ε from (1) in (2)

$$dW = -\left[-L\frac{di}{dt}\right]idt$$

$$dW = L\, i\, di$$

Total work done in establishing the current *i* is

$$W = \int dW = \int_0^i L\, i\, di = \left|\frac{i^2}{2}\right|_0^i$$

$$W = \frac{1}{2}Li^2$$

The work done is stored as magnetic potential energy.

$$U_B = \frac{1}{2}Li^2$$

28. What are the conditions for total internal reflection and derive an expression for critical angle.?

Conditions for total internal reflection to take place are,

(i) light must travel from denser to rarer medium,

(ii) angle of incidence in the denser medium must be greater than critical angle ($i > i_c$).

At critical angle $i = i_c$, $r = 90°$

Snell's law in the product form is,

$$n_1 \sin i_c = n_2 \sin 90° \qquad \sin 90° = 1$$

$$n_1 \sin i_c = n_2$$

$$\sin i_c = \frac{n_2}{n_1}$$

Here $n_2 > n_1$

For air medium, $n_2 = 1$ and the refractive index of the denser medium n_1 is taken as n

$$\sin i_c = \frac{1}{n} \text{ (or) } i_c = \sin^{-1}\left(\frac{1}{n}\right)$$

The critical angle i_c depends on the refractive index n of the medium.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

29. When light of wavelength 2200A falls on Cu, photo electrons are emitted from it. Find (i) the threshold wavelength and (ii) the stopping potential. (work function for Cu is ϕ_0=4.7eV)

i) The threshold wavelength is given by $\lambda_0 = \frac{hC}{\phi_o} = \frac{6.626\times10^{-34}\times3\times10^{8}}{4.7\times1.6\times10^{-19}}$ =2643Å

ii) Energy of the photon of wavelength 2200 Å is $E = \frac{hC}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{2200\times10^{-10}\times1.6\times10^{-19}}$ = 5.65 eV

we know the kinetic energy of the fastest photo electron is

$K_{max} = h\upsilon - \phi_0 = 5.65 - 4.7 = 0.95$ eV

$K_{max} = eV_o$

$V_o = \frac{K_{max}}{e} = \frac{0.95\times1.6\times10^{-19}}{1.6\times10^{-19}} = 0.95$ eV

30. Write a note on Hydrogen spectra.

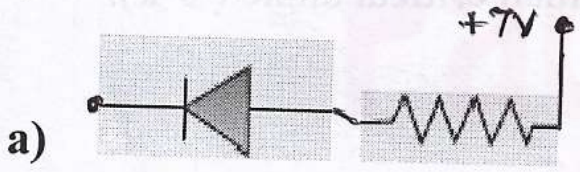
The wave number or wavelength of spectral line derived from Bohr atom model

$$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{n^2} - \frac{1}{m^2}\right)$$

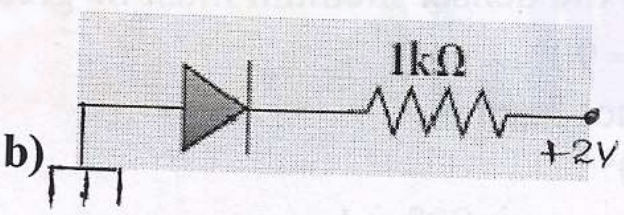
where $\bar{\nu}$ is known as wave number which is inverse of wavelength,
R is known as Rydberg constant

S.No	Spectral series	n	m	Wave Number	Region
1	Lyman series	1	2,3….	$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{m^2}\right)$	ultra-violet region
2	Balmer series	2	3,4…	$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{m^2}\right)$	Visible region
3	Paschen series	3	4,5…	$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{m^2}\right)$	Infra-red region (near IR)
4	Brackett series	4	5,6…..	$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{4^2} - \frac{1}{m^2}\right)$	Infra-red region (middle IR)
5	Pfund series	5	6,7…	$\bar{\nu} = \frac{1}{\lambda} = R\left(\frac{1}{5^2} - \frac{1}{m^2}\right)$	Infra-red region (far IR)

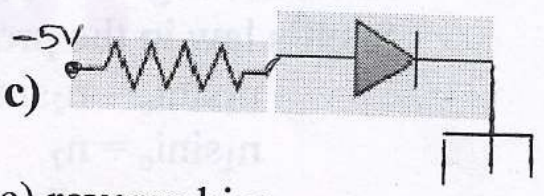
31. In the following diagram, indicate which of the diode are forward biased and which are reverse biased?



a) forward bias



b) reverse bias



c) reverse bias

32. When two resistances connected in series and parallel their equivalent resistances are 15 Ω and 56/15 Ω respectively. Find the individual resistances.

Resistors in series, the effective resistance,

$R_s = R_1 + R_2$

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

Resistors in parallel, the effective resistance,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$R_1+R_2 = 15\Omega$ ----------(1)

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{15}{56} \text{ ----------(2)}$$

$$\frac{R_1+R_2}{R_1R_2} = = \frac{15}{56}$$

substitut (1) here in Nr $\frac{15}{R_1R_2} = = \frac{15}{56}$

$R_1 R_2 = 56$ ----------(3)

From (1) $R_2 = 15 - R_1$

Substitute $R_2 = 15 - R_1$ in (3)

$R_1 (15 - R_1) = 56$

$15 R_1 - R_1^2 = 56$

$R_1^2 - 15 R_1 + 56 = 0$

$(R_1 - 8) (R_1 - 7) = 0$

$R_1 = 8$ (OR) 7

$\therefore R_1 = 8\ \Omega$

$R_2 = 7\ \Omega$

33. The light of wavelength 590nm, 596nm are used in turn to study the diffraction taking place at a single slit of aperture 2×10^{-4}m. The distance between the slit and the screen is 1.5m. Calculate the separation between the positions of first maximum of the diffraction pattern obtained in the two cases.

$\lambda_1 = 590$nm, $\lambda_2 = 596$nm

Width of the slit a= 2×10^{-6} m, D = 1.5 m

Separation between the first secondary maximum in the two cases is

$$X_2 - X_1 = 3\frac{\lambda_1 D}{2a} - 3\frac{\lambda_2 D}{2a} = \frac{3D}{2a}[\lambda_1 - \lambda_2]$$

$$= \frac{3\times1.5}{2\times2\times10^{-4}}[596 - 590]\times10^{-9}$$

$$= \frac{3\times1.5\times10^{-9}}{2\times2\times10^{-4}}$$

$$= 6.75\times10^{-5}\ \text{m}$$

PART - IV

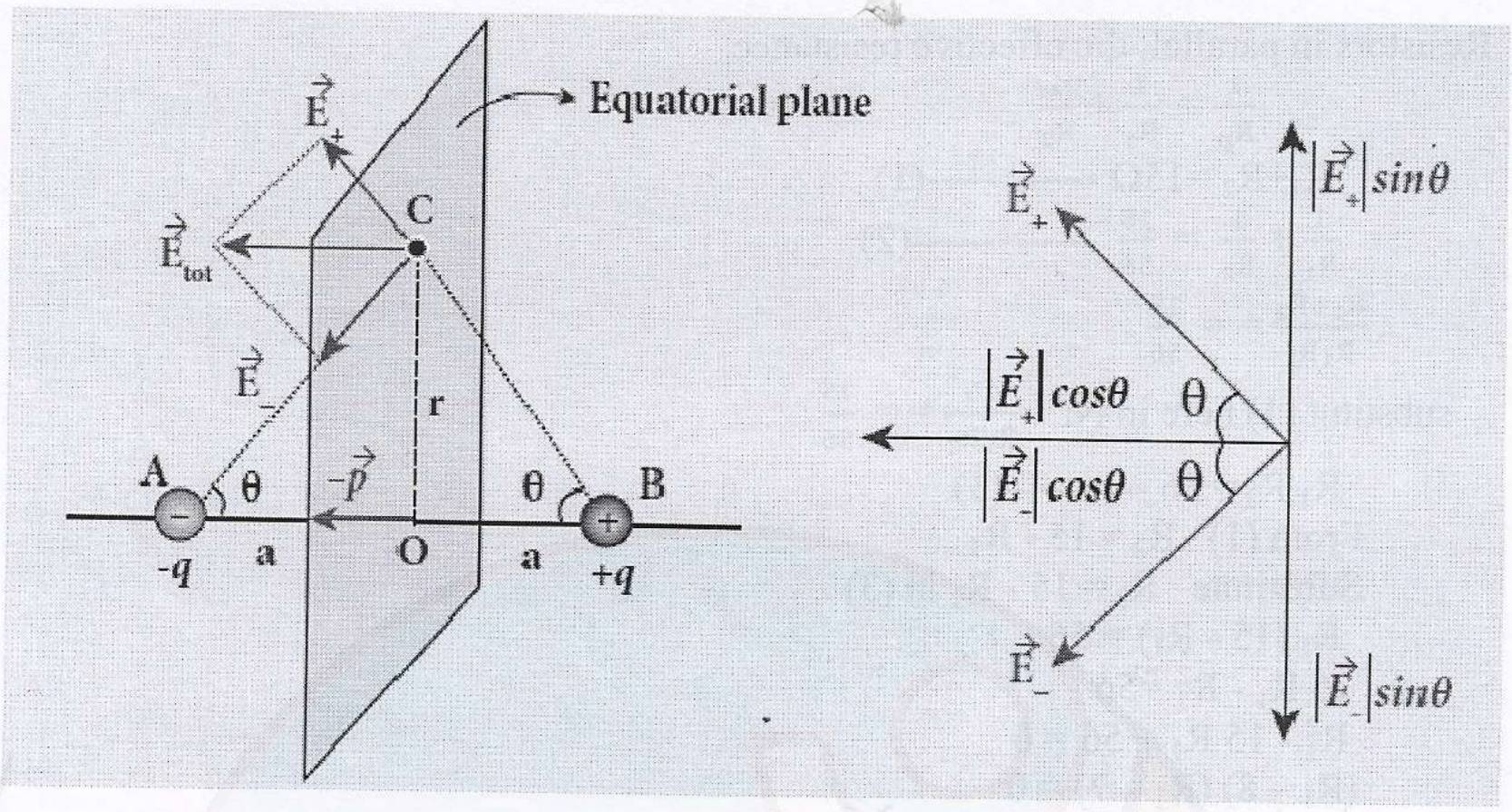
Answer all the questions $5 \times 5 = 25$

34. (a) What is electric dipole? Derive an expression for the electric field at a point due to an electric dipole along the equatorial plane.

Consider a point C at a distance r from the midpoint O of the dipole on the equatorial plane as shown in Figure

Since the point C is equi-distant from +q and –q, the magnitude of the electric fields of +q and –q are the same. The direction of $\vec{E}_+$ is along BC and the direction of $\vec{E}_-$ is along CA. $\vec{E}_+$ and $\vec{E}_-$ are resolved into two components; one component parallel to the dipole axis and the other perpendicular to it.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com



The perpendicular components $|\vec{E}_+|\sin\theta$ and $|\vec{E}_-|\sin\theta$ are oppositely directed and cancel each other.

The magnitude of the total electric field at point C is the sum of the parallel components of $|\vec{E}_+|\cos\theta$ and $|\vec{E}_-|\cos\theta$ and its direction is along $-\hat{p}$ as shown in the Figure

$$\vec{E}_{tot} = -|\vec{E}_+|\cos\theta\hat{p} - |\vec{E}_-|\cos\theta\,\hat{p} \quad \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

The magnitudes $\vec{E}_+$ and $\vec{E}_-$ are the same and are given by

$$|\vec{E}_+| = |\vec{E}_-| = \frac{1}{4\pi\varepsilon_o} q\left(\frac{q}{(r^2+a^2)}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

By substituting equation (2) into equation (1), we get

$$\vec{E}_{tot} = -2|\vec{E}_+|\cos\theta$$

$$= -\frac{1}{4\pi\varepsilon_o}\left(\frac{2q\cos\theta}{(r^2+a^2)}\right)\hat{p} \quad \ldots\ldots\ldots\ldots\ldots(3)$$

In $\triangle OBC$ $\cos\theta = \frac{a}{\sqrt{r^2+a^2}}$

Substitute $\cos\theta$ in the above equation(3)

$$\vec{E}_{tot} = -\frac{1}{4\pi\varepsilon_o}\left(\frac{2q}{(r^2+a^2)}\right)\frac{a}{\sqrt{r^2+a^2}}\hat{p}$$

$$\vec{E}_{tot} = -\frac{1}{4\pi\varepsilon_o}\frac{\vec{p}}{(r^2+a^2)^{\frac{3}{2}}} \quad \ldots\ldots\ldots\ldots\ldots(4)$$

Since $2aq\hat{p} = \vec{p}$

At very large distances ($r >> a$), the equation (4) becomes

$$\vec{E}_{tot} = \frac{1}{4\pi\varepsilon_o}\frac{\vec{p}}{r^3} \quad (r >> a) \ldots\ldots(5)$$

The direction of the electric field at points on the equatorial plane it is directed opposite to the dipole moment vector, that is along $-\vec{p}$

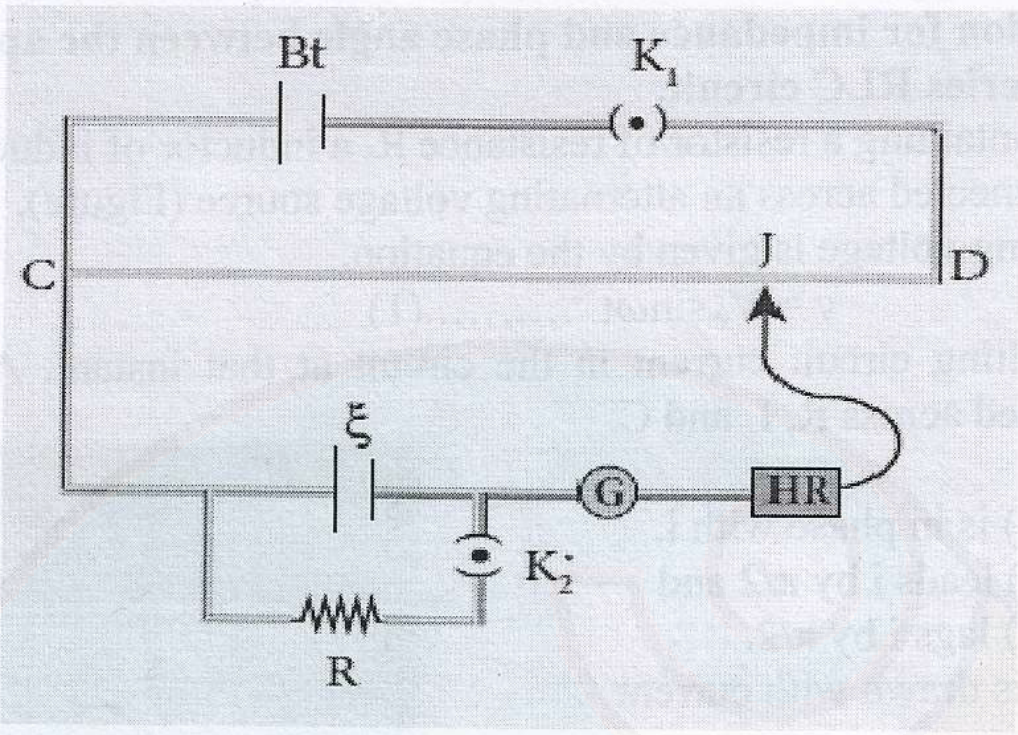
(OR)

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

b) i) Explain the determination of the internal resistance of a cell using a potentiometer.

To measure the internal resistance of a cell, the circuit connections are made as shown in Figure



The end C of the potentiometer wire is connected to the positive terminal of the battery Bt and the negative terminal of the battery is connected to the end D through a key K_1. This forms the **primary circuit.**

The positive terminal of the cell ξ whose internal resistance is to be determined is also connected to the end C of the wire. The negative terminal of the cell ξ is connected to a jockey through a galvanometer and a high resistance. This is the **secondary circuit**

A resistance box R and key K_2 are connected across the cell ξ.

With K_2 open, the balancing point J is obtained and the balancing length CJ = l_1 is measured. Since the cell is in open circuit, its emf is

$$\xi \propto l_1$$

A suitable resistance (say, 10 Ω) is included in the resistance box and key K_2 is closed. Let r be the internal resistance of the cell. The current passing through the cell and the resistance R is given by

$$I = \frac{\xi}{R+r} \quad \text{----------(1)}$$

Hence the potential difference across R = V = $\frac{\xi R}{R+r}$ ----------(2)

When this potential difference is balanced on the potentiometer wire, let l_2 be the balancing length. Then $\frac{\xi R}{R+r} \propto l_2$ ----------(3)

From equations (2) and (3) $\frac{R+r}{R} = \frac{l_1}{l_2}$

$$1+\frac{r}{R} = \frac{l_1}{l_2}$$

$$r = R[\frac{l_1}{l_2} - 1]$$

Substituting the values of the R, l_1 and l_2 , the internal resistance of the cell is determined. The experiment can be repeated for different values of R. It is found that the internal resistance of the cell is not constant but increases with increase of external resistance connected across its terminals.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

ii) Why is a potentiometer preferred over a voltmeter for comparison of emf's of cells?

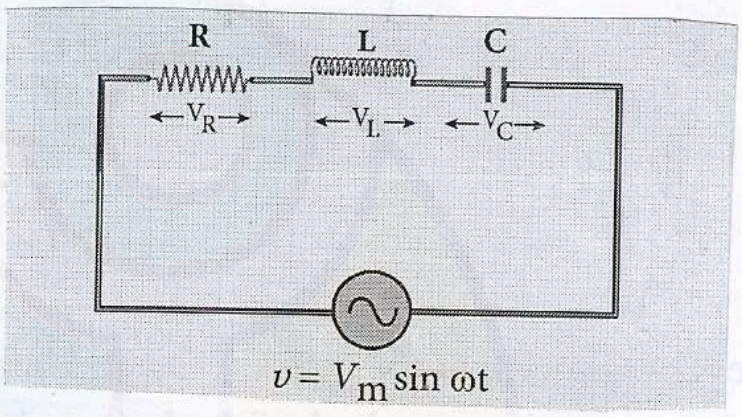
Voltmeter always draw some current from the cell,
potentiometer does not draw any current for itself from the primary cell.
Hence potentiometer is preferred over a voltmeter for comparison of emf's of cells

35. a) Derive an expression for impedance and phase angle between the applied voltage and current in a series RLC circuit.

Consider a circuit containing a resistor of resistance R, a inductor of inductance L and a capacitor of capacitance C connected across an alternating voltage source (Figure).
The applied alternating voltage is given by the equation.

$$v = V_m \sin\omega t \ \ldots\ldots\ldots(1)$$

Let *i* be the resulting circuit current in the circuit at that instant. As a result, the voltage is developed across R, L and C.

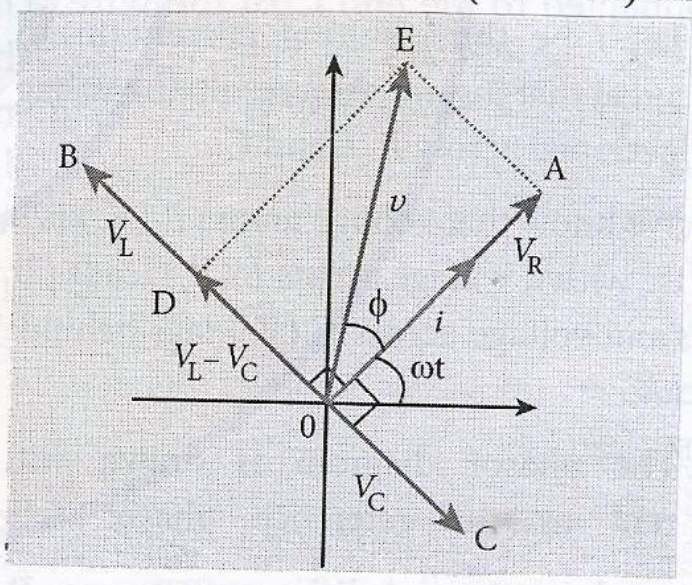


We know that
voltage across R (V_R) is in phase with i,
voltage across L (V_L) leads i by $\pi/2$ and
voltage across C (V_C) lags i by $\pi/2$.
The phasor diagram is drawn with current as the reference phasor.

The current *i* is represented by the phasor $\overrightarrow{OI}$, V_R by $\overrightarrow{OA}$, V_L by $\overrightarrow{OB}$ and V_C by $\overrightarrow{OC}$ as shown in Figure

The net voltage drop across L-C combination is $V_L - V_C$ which is represented by a phasor $\overrightarrow{OD}$

By parallelogram law, the diagonal $\overrightarrow{OE}$ gives the resultant voltage *v* of V_R and $(V_L - V_C)$ and its length OE is equal to V_m. Therefore,



$$V_m^2 = V_R^2 + (V_L - V_C)^2$$

The effective voltage(V_m)

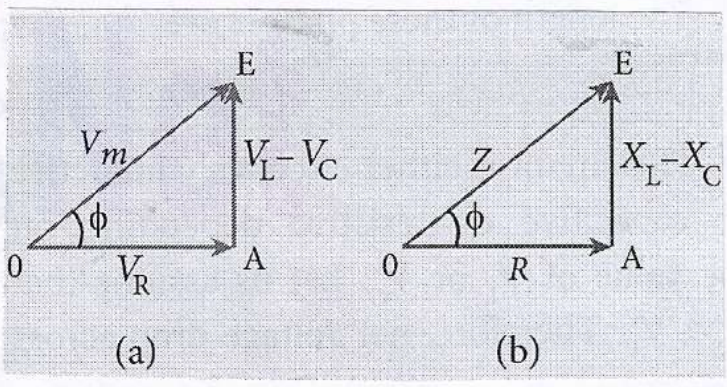
$$V_m = \sqrt{(I_m R)^2 + (I_m X_L - I_m X_C)^2}$$

$$= I_m \sqrt{R^2 + (X_L - X_C)^2}$$

$$\text{(or) } I_m = \frac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\text{(or) } I_m = \frac{V_m}{Z}$$

Where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ = Impedance of the circuit
The voltage triangle and impedance triangle are given in the Figure



From phasor diagram, the phase angle between *v* and *i* is found out from the following relation

$$\tan\phi = \left(\frac{V_L - V_C}{R}\right) = \left(\frac{I_m X_L - I_m X_C}{I_m R}\right) = \left(\frac{X_L - X_C}{R}\right)$$

Special cases

(i) If $X_L > X_C$, $(X_L - X_C)$ is positive and phase angle ϕ is also positive. It means that the applied voltage leads the current by ϕ (or current lags behind voltage by ϕ). The circuit is inductive.

$$\therefore v = V_m \sin\omega t \ , \ i = I_m \sin(\omega t - \phi)$$

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

(ii) If $X_L < X_C$, $(X_L - X_C)$ is negative and ϕ is also negative. Therefore current leads voltage by ϕ and the circuit is capacitive.

$\therefore v = V_m \sin\omega t \ , i = I_m \sin(\omega t + \phi)$

(iii) If $X_L = X_C$, ϕ is zero. Therefore current and voltage are in the same phase and the circuit is resistive.

$\therefore v = V_m \sin\omega t \ , i = I_m \sin\omega t$

(OR)

b) Give the advantages and limitations of frequency modulation. Why a phase modulation provides high transmission speed?

a) Advantages of FM

i) In FM, there is a large decrease in noise.

ii) The operating range is quite large.

iii) The transmission efficiency is very high

iv) FM bandwidth covers the entire frequency range.

Due to this, FM radio has better quality compared to AM radio.

b) Limitations of FM

i) FM requires a much wider channel.

ii) FM transmitters and receivers are more complex and costly.

iii) In FM reception, less area is covered compared to AM.

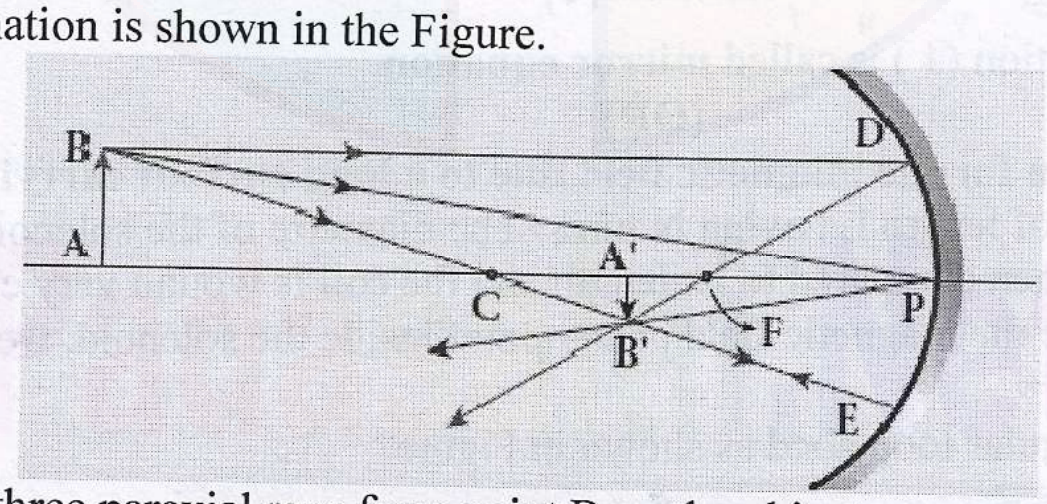
c) The waveform of the phase modulated wave is similar to that of FM wave. The phase modulation, generally uses a smaller bandwidth than FM. In other words, more information can be sent in a given bandwidth in phase modulation. Therefore, phase modulation facilitates highest transmission speeds on a given bandwidth.

36. a) Derive an expression for mirror equation

The mirror equation establishes a relation among object distance *u*, image distance *v* and focal length *f* for a spherical mirror.

An object AB is considered on the principal axis of a concave mirror beyond the center of curvature C.

The image formation is shown in the Figure.



Let us consider three paraxial rays from point B on the object.

The first paraxial ray BD travelling parallel to principal axis is incident on the concave mirror at D, close to the pole P. After reflection the ray passes through the focus F.

The second paraxial ray BP incidentat the pole P is reflected along PB′.

The third paraxial ray BC passing through centre of curvature C, falls normally on the mirror at E is reflected back along the same path.

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

The three reflected rays intersect at the point A′B′. A perpendicular drawn as A′B′ to the principal axis is the real, inverted image of the object AB.

As per law of reflection, the angle of incidence ∠ BPA is equal to the angle of reflection ∠B′PA′ .

The triangles ΔBPA and ΔB′PA′ a similar. Thus, from the rule of similar triangles,

$$\frac{A'B'}{AB} = \frac{PA'}{PA} \quad \text{.....(1)}$$

The other set of similar triangles are, ΔDPF and Δ B′A′ F. (PD is almost a straight vertical line)

$$\frac{A'B'}{PD} = \frac{A'F}{PF}$$

As, the distances PD = AB the above equation becomes,

$$\frac{A'B'}{AB} = \frac{A'F}{PF} \quad \text{----------(2)}$$

From equations (1) and (2) we can write,

As, A′ F = PA′ - PF , the above equation becomes,

$$\frac{PA'}{PA} = \frac{PA' - PF}{PF} \quad \text{---------(3)}$$

We can apply the sign conventions for the various distances in the above equation.

PA= −u, PA' = −v, PF = −f

All the three distances are negative as per sign convention, because they are measured to the left of the pole. Now, the equation (3) becomes,

$$\frac{-v}{-u} = \frac{-v-(-f)}{-f} \quad ; \quad \frac{v}{u} = \frac{v}{f} - 1$$

On further simplification,

$$\frac{v}{u} = \frac{v-f}{f} \quad ; \quad \frac{v}{u} = \frac{v}{f} - 1$$

Dividing either side with v, $\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$

After rearranging, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ ----------(4)

The above equation (4) is called **mirror equation**.

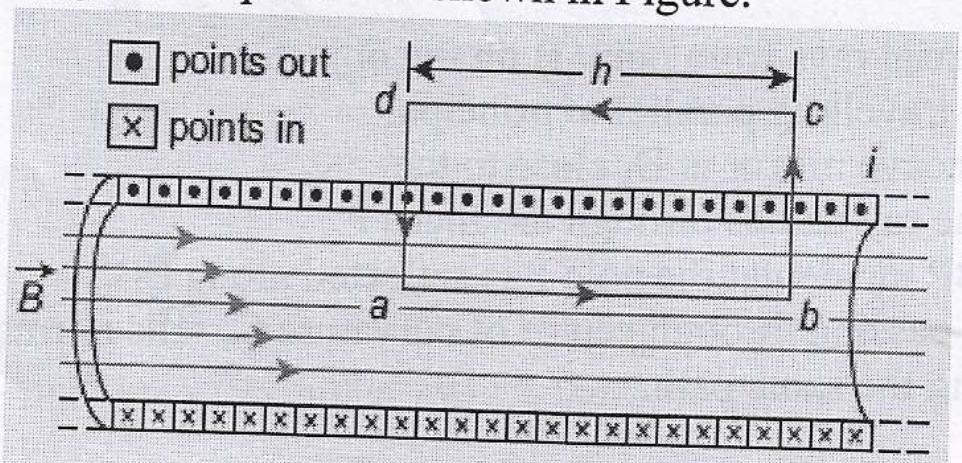
(OR)

b). Derive an expression for the magnetic field due to a long current carrying solenoid

Consider a solenoid of length L having N turns. The diametre of the solenoid is assumed to be much smaller when compared to its length and the coil is wound very closely.

In order to calculate the magnetic field at any point inside the solenoid, we use Ampere's circuital law.

Consider a rectangular loop abcd as shown in Figure.



Magnetic field of a solenoid

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Then from Ampere's circuital law,

$$\oint_c \vec{B}.d\vec{l} = \mu_o I_{enclosed}$$

$$= \mu_o \times \text{(total current enclosed by Amperian loop)}$$

The left hand side of he equation is

$$\oint_c \vec{B}.d\vec{l} = \int_a^b \vec{B}.d\vec{l} + \int_b^c \vec{B}.d\vec{l} + \int_c^d \vec{B}.d\vec{l} + \int_d^a \vec{B}.d\vec{l}$$

The elemental lengths along bc and da are perpendicular to the magnetic field

$$\int_b^c \vec{B}.d\vec{l} = \int_b^c |\vec{B}|\,|d\vec{l}| \cos 90° = 0$$

$$\int_d^a \vec{B}.d\vec{l} = 0$$

The magnetic field outside the solenoid is zero , for the path cd , the integral is

$$\int_c^d \vec{B}.d\vec{l} = 0$$

For the path along ab, the integral is

$$\int_a^b \vec{B}.d\vec{l} = B\int_a^b dl \cos 0° = B\int_a^b dl = BL$$

where L is the length of tyhe solenoid

Let NI be the current passing through the solenoid of N turns, the

$$\int_a^b \vec{B}.d\vec{l} = BL = \mu_o NI$$

$$\therefore B = \mu_o \frac{NI}{L}$$

If n be the number of turns per unit length, $n = \frac{N}{L}$

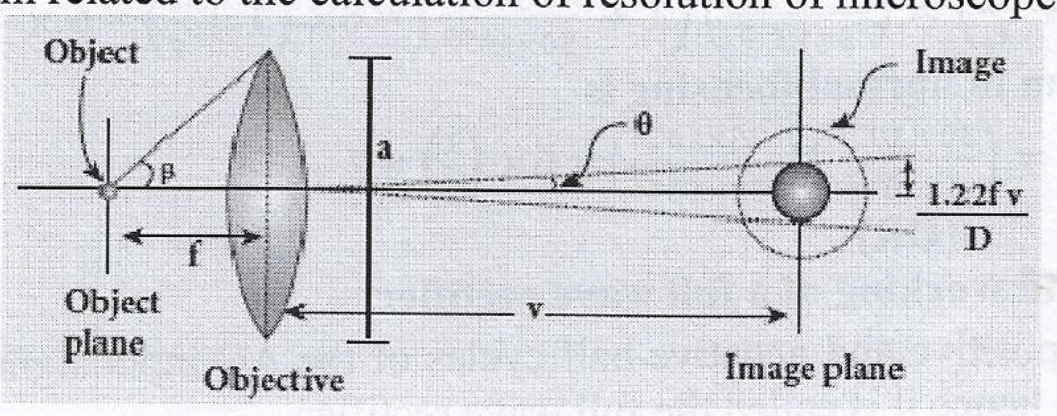
Hence $B = \mu_o nI$

Since *n* is a constant for a given solenoid and μ_0 is also constant.

For a fixed current I, the magnetic field inside the solenoid is also a constant

37 a). i) Obtain the expression for resolving power of a microscope.

The diagram related to the calculation of resolution of microscope is illustrated in Figure



Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

A microscope is used to see the details of the object under observation. The ability of microscope depends not only in magnifying the object but also in resolving two points on the object separated by a small distance d_{min}. Smaller the value of d_{min} better will be the resolving power of the microscope.

The radius of central maxima is already derived as,

$$r_o = \frac{1.22\lambda v}{a}$$

In the place of focal length f we have the image distance v.

If the difference between the two points on the object to be resolved is d_{min},

$$\text{magnification } m = \frac{r_o}{d_{min}}$$

$$d_{min} = \frac{r_o}{m}$$

$$= \frac{1.22\lambda v}{am} = \frac{1.22\lambda v}{a\left(\frac{v}{u}\right)}$$

$$= \frac{1.22\lambda u}{a} \quad [\because m = v/u]$$

$$d_{min} = \frac{1.22\lambda f}{a} \quad [u \approx f]$$

On the object side,

$$2\tan\beta = 2\sin\beta = \frac{a}{f} \quad \left[\tan\beta = \frac{\left(\frac{a}{2}\right)}{f} \quad \therefore a = f2\sin\beta\right]$$

$$d_{min} = \frac{1.22\lambda}{2\sin\beta}$$

To further reduce the value of d_{min} the optical path of the light is increased by immersing the objective of the microscope in to a bath containing oil of refractive index n.

$$d_{min} = \frac{1.22\lambda}{2n\sin\beta}$$

Such an objective is called oil immersed objective. **The term nsinβ is called numerical aperture NA.**

$$d_{min} = \frac{1.22\lambda}{2\,(NA)}$$

ii) A microscope has an objective and eyepiece of focal lengths 5 cm and 50 cm respectively with tube length 30 cm. Find the magnification of the microscope in the (i) near point and (ii) normal focusing

.Given : $f_o = 5\text{ cm} = 5 \times 10^{-2}$ m, $f_e = 50\text{ cm} = 50 \times 10^{-2}$ m

$L = 30\text{ cm} = 30 \times 10^{-2}$ m, $D = 25\text{ cm} = 25 \times 10^{-2}$ m

i) **Total magnification m in near point focusing is**

$$m = m_o m_e = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right) = \left(\frac{30\times10^{-2}}{5\times10^{-2}}\right)\left(1 + \frac{25\times10^{-2}}{50\times10^{-2}}\right) = 6\times\left(1 + \frac{1}{2}\right) = 6 \times \frac{3}{2} = 9$$

ii) **Total magnification m in normal focusing is**

$$m = m_o m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right) = \left(\frac{30\times10^{-2}}{5\times10^{-2}}\right)\left(\frac{25\times10^{-2}}{50\times10^{-2}}\right) = 6\times\left(\frac{1}{2}\right) = 3$$

(OR)

b). Explain the construction and working of a full wave rectifier.

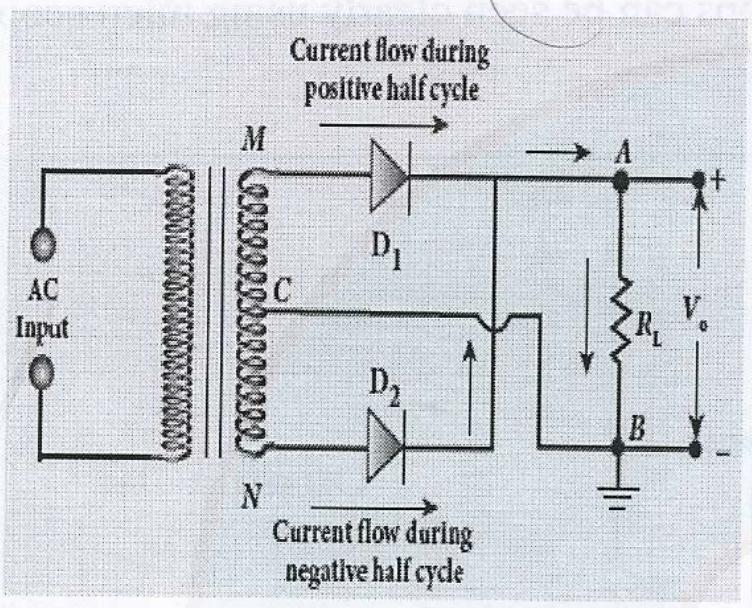
Full wave rectifier: The positive and negative half cycles of the AC input signal are rectified in this circuit and hence it is called the full wave rectifier.

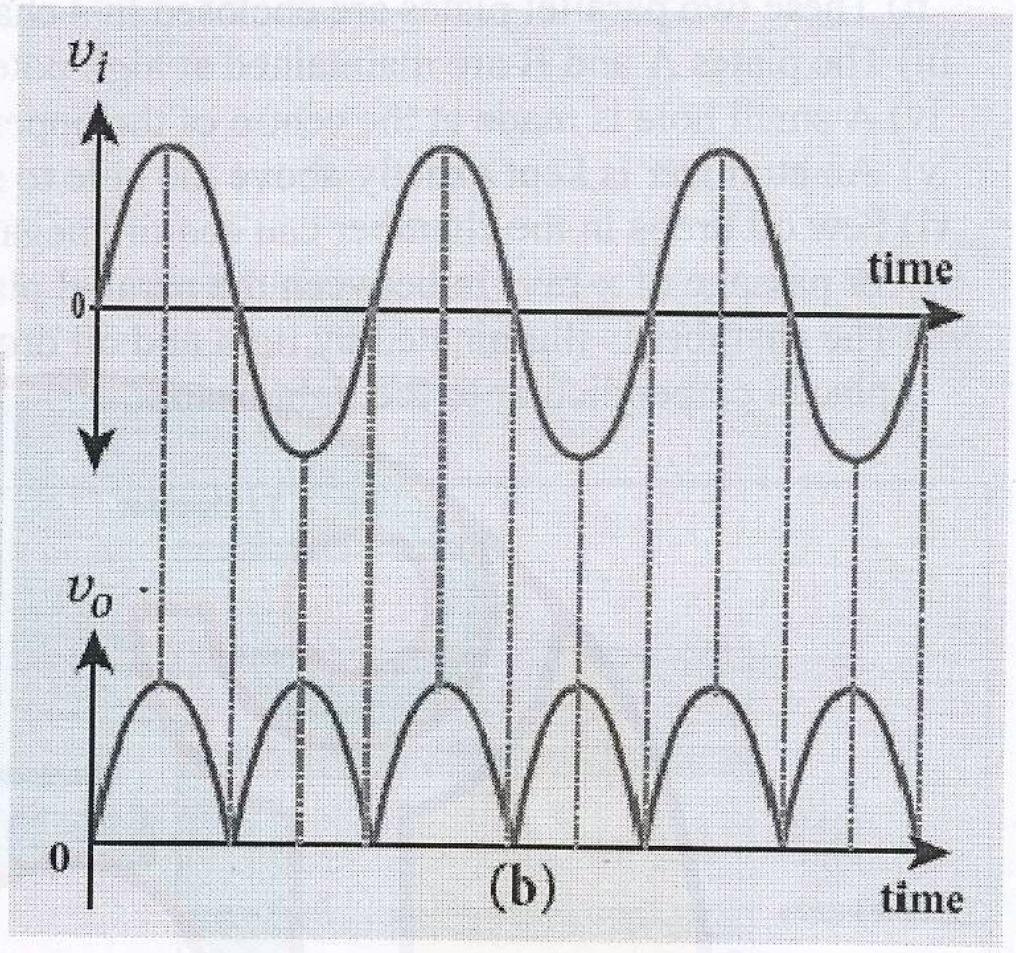
Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

Circuit:

It consists of two p-n junction diodes, a centre tap transformer and a load resistor R_L. The centre is usually taken as the ground or zero voltage reference point





Input signal	Terminal M	Centre tap C	Terminal M	Diode D_1	Diode D_2	path of the current flow	Output voltage
During the positive half cycle of the AC	Positive	Zero	Negative	Forward bias- Conducts	Reverse bias- do not conducts	$MD_1A(R_L)BC$	V_0
During the negative half cycle of the AC	Negative	Zero	Positive	Reverse bias- do not conduct	Forward bias- conduct	$ND_2A(R_L)BC$	V_0

During both positive and negative half cycles of the input signal, the current flows through the **load resistance R_L in the same direction.**

Here both half cycles of AC input are rectified, the output is still pulsating in nature.

Efficiency (η) $= \frac{\text{output DC power}}{\text{input AC power}} \times 100 = 81.2\ \%$ **(twice of a half wave rectifier)**

38. a) Discuss the Millikan's oil drop experiment to determine the charge of an electron.

Millikan's oil drop experiment is used to determine as charge of an electron

Principle:

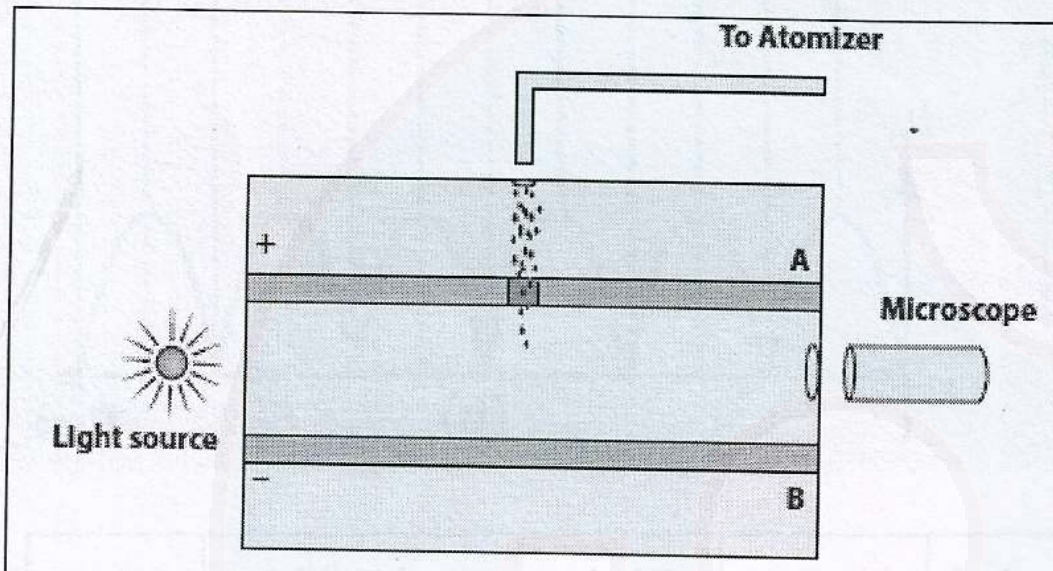
By adjusting electric field suitably, the motion of oil drop inside the chamber can be controlled – that is, it can be made to move up or down or even kept balanced in the field of view for sufficiently long time.

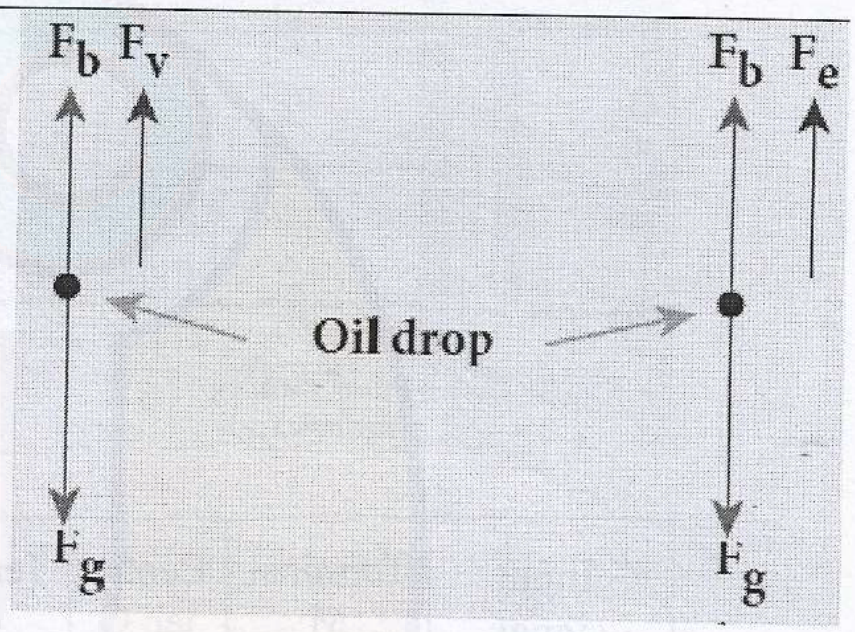
Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

Construction:

i) The apparatus consists of two horizontal circular metal plates A and B each with diameter around 20 cm and are separated by a small distance 1.5 cm.
ii) These two parallel plates are enclosed in a chamber with glass walls.
iii) The plates A and B are maintained at high potential difference around 10 kV.
iv) A small hole is made at the centre of the upper plate A
v) An atomizer is kept exactly above the hole to spray the highly viscous the liquid.
vi) Few oil drops in the chamber can acquire negative charge because of friction with air or passage of x-rays in between the parallel plates.
vii) The chamber is illuminated by light and oil drops can be seen clearly using microscope placed perpendicular to the light beam.





Let m be the mass of the oil drop and q be its charge. Then the forces acting on the droplet are

(a) gravitational force $F_g = mg$
(b) electric force $F_e = qE$
(c) buoyant force F_b
(d) viscous force Fv

(a) Determination of radius of the droplet

When the electric field is switched off, the oil drop accelerates downwards.
we note that viscous force and buoyant force balance the gravitational force.
Once the oil drop attains a terminal velocity *v*, the net downward force acting on the oil drop is equal to the viscous force acting opposite to the direction of motion of the oil drop.
From the free body diagram(a)

$$F_g = F_b + F_v$$

$$\rho\left[\frac{4}{3}\pi r^3\right]g = \sigma\left[\frac{4}{3}\pi r^3\right]g + 6\pi r v\eta$$

Let ρ - the density of the oil drop; σ - the density of air

v- terminal velocity and r -the radius of the oil drop

$$\frac{4}{3}\pi r^3(\rho - \sigma)\, g = 6\pi r v\eta$$

$$\therefore \quad r = \left[\frac{9\eta v}{2(\rho-\sigma)g}\right]^{\frac{1}{2}} \text{ ----------(1)}$$

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

Padasalai.Net CBSEtips.in

(b) Determination of electric charge

When the electric field is switched on, charged oil drops experience an upward electric force (qE).

If a drop is stationary, there will be no viscous force acting on the oil drop.

From the free body diagram (b)

$$F_e + F_b = F_g$$

$$qE + \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 \rho g$$

$$qE = \frac{4}{3}\pi r^3 (\rho - \sigma) g$$

$$qE = \frac{4}{3E}\pi r^3 (\rho - \sigma) g \quad \text{----------(2)}$$

substitute (1) in (2)

$$q = \frac{18\pi}{E}\left[\frac{\eta^3 v^3}{2(\rho-\sigma)g}\right]^{\frac{1}{2}}$$

Charge of the electron – 1.6×10^{-19} C

(OR)

38. **b) Discuss the properties of dia ,para and ferro magnetic materials**

S No	properties	Dia magnetic material	Para magnetic material	Ferro magnetic material
1	The net magnetic moment	zero	zero	It possesses some value.
2	In the presence of an external magnetic field,	there is net magnetic dipole moment induced in the opposite direction to the applied field.	there is net magnetic dipole moment induced in the direction of the applied field.	In the presence of external magnetic field, (i) the domains having magnetic moments parallel to the field grow in size (ii) the other domains (not parallel to field) are rotated so that they are aligned with the field.
3	When placed in a non-uniform magnetic field	the material tends to move from **stronger part to weaker part** o the external field.	the materials will have a tendency to move from **weaker to stronger** part of the field.	the materials will have a strong tendency to move from **weaker to stronger** part of the field.
4	Magnetic susceptibility	**negative**	**positive and small.**	**positive and large.**
5	Relative permeability	slightly **less than unity**	**greater than unity.**	**large.**
6	The magnetic field lines	are **repelled or expelle** by diamagnetic materia when placed in a magn field. iv)	are **attracted** into the paramagnetic materials when placed in a magnetic field.	are strongly **attracted** into the ferromagnetic materials when placed in a magnetic field.
7	Susceptibility	is nearly **temperature independent**	is **inversely proportional to temperature**	is **inversely proportional to temperature**

T. JOHN PETER
9566551159

Please Send Your Model Questions to our Email Id: padasalai.net@gmail.com

www.Padasalai.Net

படங்களை தொடுக! பாடசாலை வலைதளத்தை சமூக ஊடகங்களில் பின்தொடர்க!! உடனுக்குடன் புதிய செய்திகளை Notifications-ல் பெறுக!

Zoom Touch Below Links Download!

12th Standard	Syllabus	Books	Study Materials – EM	Study Materials - TM	Practical	Online Test (EM & TM)
	Monthly Q&A	Mid Term Q&A	Revision Q&A	PTA Book Q&A	Centum Questions	Creative Questions
	Quarterly Exam	Half Yearly Exam	Public Exam	NEET		

11th Standard	Syllabus	Books	Study Materials – EM	Study Materials - TM	Practical	Online Test (EM & TM)
	Monthly Q&A	Mid Term Q&A	Revision Q&A	Centum Questions	Creative Questions	
	Quarterly Exam	Half Yearly Exam	Public Exam	NEET		

10th Standard	Syllabus	Books	Study Materials - EM	Study Materials - TM	Practical	Online Test (EM & TM)
	Monthly Q&A	Mid Term Q&A	Revision Q&A	PTA Book Q&A	Centum Questions	Creative Questions
	Quarterly Exam	Half Yearly Exam	Public Exam	NTSE	SLAS	

9th Standard	Syllabus	Books	Study Materials	1st Mid Term	2nd Mid Term	3rd Mid Term
	Quarterly Exam	Half Yearly Exam	Annual Exam	RTE		

8th Standard	Syllabus	Books	Study Materials	1st Mid Term	2nd Mid Term	3rd Mid Term
	Term 1	Term 2	Term 3	Public Model Q&A	NMMS	Periodical Test

7th Standard	Syllabus	Books	Study Materials	1st Mid Term	2nd Mid Term	3rd Mid Term
	Term 1	Term 2	Term 3	Periodical Test	SLAS	

6th Standard	Syllabus	Books	Study Materials	1st Mid Term	2nd Mid Term	3rd Mid Term
	Term 1	Term 2	Term 3	Periodical Test	SLAS	

1st to 5th Standard	Syllabus	Books	Study Materials	Periodical Test	SLAS	
	Term 1	Term 2	Term 3	Public Model Q&A		

Exams	TET	TNPSC	PGTRB	Polytechnic	Police	Computer Instructor
	DEO	BEO	LAB Asst	NMMS	RTE	NTSE

Portal	Matrimony	Mutual Transfer	Job Portal

Volunteers	Centum Team	Creative Team	Key Answer Team

Downloads	LESSON PLAN	Department Exam	Income Tax	Forms & Proposals	Fonts	Downloads
	Proceedings	GO's	Regulation Orders	Pay Orders	Panel	

Padasalai – Official Android App – Download Here

Kindly Send Your Study Materials, Q&A to our Email ID – Padasalai.net@gmail.com